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$$\int \frac{x^2 - x}{(x^2 + 1)(x + 1)^2} dx = \frac{Ax + B}{x^2 + 1} + \frac{C}{(x + 1)^2} + \frac{D}{x + 1}$$

$$C = \frac{x^2 - x}{(x^2 + 1)} \Big|_{x=-1} = 1$$

$$Ax + B = \frac{-1 - i}{(i + 1)^2} = \frac{-1}{2} + \frac{i}{2}$$

$$\Rightarrow B = \frac{-1}{2} \text{ en } A = \frac{1}{2}$$

$$\frac{x^2 - x}{(x^2 + 1)(x + 1)^2} - \frac{1}{(x + 1)^2} = \frac{-1}{(x^2 + 1)(x + 1)}$$

$$\Rightarrow D = \frac{-1}{(x^2 + 1)} \Big|_{x=-1} = \frac{-1}{2}$$

$$\Rightarrow \int \frac{\frac{1}{2}x - \frac{1}{2}}{x^2 + 1} dx + \int \frac{dx}{(x + 1)^2} - \frac{1}{2} \int \frac{1}{x + 1} dx$$

$$= \int \frac{\frac{1}{2}x}{x^2 + 1} dx - \frac{1}{2} \text{Bgtan } x + \int (x + 1)^{-2} dx - \frac{1}{2} \ln |x + 1|$$

$$= \frac{1}{4} \ln |x^2 + 1| - \frac{1}{2} \text{Bgtan } x - \frac{1}{x + 1} - \frac{1}{2} \ln |x + 1| + C$$

References